

On Modality, Existential Graphs, and the Unenclosable Sheet

A conversation · excluded middle, the conditional, the Endoporeutic Game, and a rendering of the Tao Te Ching

MICHAEL

What are the principles of excluded middle and contradiction in regard to propositions?

CLAUDE

The two are classical “laws of thought,” and the key to understanding them is that they make *different* claims that only coincide under bivalence.

Principle of (non-)contradiction. No proposition is both true and false. A proposition and its negation cannot both hold at once: formally, $\neg(P \wedge \neg P)$, or equivalently “of P and not-P, *at most one* is true.” It is the constraint that forbids a proposition from being affirmed and denied in the same respect at the same time.

Principle of excluded middle. Every proposition is either true or false, with no third value: $P \vee \neg P$, or “of P and not-P, *at least one* is true.” It forbids a gap between a proposition and its negation.

The clean way to keep them apart: contradiction says *not both*, excluded middle says *not neither*. Non-contradiction rules out **overlap** (gluts); excluded middle rules out **gaps**. Conjoin them and you get strict bivalence — exactly one of P, $\neg P$ holds. Drop one and you get the familiar non-classical families: paraconsistent logics relax non-contradiction (tolerating true contradictions), while intuitionistic logic rejects excluded middle as a general law (it won’t grant $P \vee \neg P$ without a construction deciding which disjunct holds, even though it *retains* non-contradiction).

Given your Peircean interests, the part worth flagging is that Peirce treated these two not as a symmetric pair but as the two distinct ways a sign can be *indeterminate*. His formulation (around CP 5.447–448) ties each law to one mode of indeterminacy:

- A sign is **general** insofar as the principle of excluded middle does *not* apply to it. With a genuine generality, the right to pick out instances is reserved to the interpreter, so the sign as such is committed neither to P nor to $\neg P$ of any given case — the “middle” is open. (“Man is mortal” doesn’t determine *which* man.)
- A sign is **vague** (indefinite) insofar as the principle of contradiction does *not* apply to it. Here the utterer reserves the right of further determination, so both a predicate and its denial may be loosely assertable until the sign is pinned down — the prohibition on “both” is suspended.

So for Peirce, a fully *determinate* proposition is one to which both laws apply (the bivalent case), and generality and vagueness are precisely the controlled relaxations of

excluded middle and contradiction respectively. That mapping — interpreter-side indeterminacy $\rightarrow$ excluded middle, utterer-side indeterminacy $\rightarrow$ contradiction — is one of the more elegant moves in his account, and it's why he could hold that the two laws have genuinely independent scope rather than being a single "law of bivalence."

Want me to work through how this plays out in the Existential Graphs, where negation is the cut and excluded-middle/contradiction show up as facts about the empty cut and double cuts on the Sheet of Assertion?

MICHAEL

When one says, "if P then Q," does that entail a possible Q conditioned on P?

CLAUDE

Short answer: not on the standard reading of "if P then Q." The intuition that it should is real and principled, but it's exactly the intuition that the material conditional fails to honor — which is why it pushes you toward modal, relevance, or connexive systems, and on your side of the fence toward Peirce's distinction between the conditional *de inesse* and the would-be.

Let me separate the modal targets, because "a possible Q conditioned on P" can mean either $\Diamond Q$ (Q is possible *simpliciter*) or $\Diamond(P \wedge Q)$ (P and Q are *jointly* possible — Q is a live option under P). The material conditional entails neither.

$P \rightarrow Q$ is just $\neg P \vee Q$. It makes no modal claim whatever; it reports the actual distribution of truth-values and nothing about what is possible. The decisive counterexamples are the vacuous cases:

- Let Q be a contradiction ($R \wedge \neg R$), hence impossible, and let P be false. Then $P \rightarrow Q$ is *true* (the antecedent fails), yet $\neg \Diamond Q$. So a true conditional with a flatly impossible consequent — refuting " $P \rightarrow Q \models \Diamond Q$."
- Let P itself be impossible. Then $P \rightarrow Q$ is true for *every* Q, while $\Diamond(P \wedge Q)$ is false because P can't obtain. So no joint possibility either — and worse, $P \rightarrow Q$ and $P \rightarrow \neg Q$ hold *together*. Whenever the antecedent is impossible (or merely false), the conditional carries zero information about Q's standing under P.

That last point is the heart of it: the very thing your question wants — that asserting "if P then Q" should rule out "if P then not-Q," so that Q is a genuine option under P — is *Boethius' thesis*, and it is the defining commitment of connexive logic precisely because classical logic denies it. The same dissatisfaction motivates relevance logic, which refuses to let an irrelevant or impossible P license Q.

Now, when *does* the entailment go through? Tighten the conditional to strict implication, $\Box(P \rightarrow Q)$, and add that the antecedent is genuinely possible, $\Diamond P$. Then it follows that $\Diamond(P \wedge Q)$: some accessible world has P; in that world $P \rightarrow Q$ holds; so Q holds

there; so P-and-Q are jointly possible. The load-bearing premise is $\Diamond P$. Strict implication without it is just as vacuous as the material conditional — an impossible antecedent makes $\Box(P \rightarrow Q)$ trivially true and yields nothing.

This is essentially Peirce's contrast. The conditional *de inesse* speaks only of the actual occasion ("it is not the case that P-and-not-Q, here, now") and is silent on possibility — it behaves exactly like the material conditional and inherits its vacuities. The substantive conditional for Peirce is the *would-be*: a real, law-like disposition grounded in his realism about possibility (Thirdness). And note what the would-be actually asserts in the consequent — not that Q is merely *possible* given P, but that Q *would* obtain were P realized, which is a necessity-relative-to-P. The "possible Q conditioned on P" you're after sits one notch weaker than that: it's what you get as a *presupposition* of the would-be being non-idle, namely that P is a real possibility and doesn't foreclose Q. Strip away the reality of the antecedent's possibility and the would-be collapses back into an empty *de inesse* form.

So: "if P then Q" entails a genuinely possible Q-under-P only when (a) the conditional is read as law-like / strict rather than *de inesse*, and (b) P is independently a real possibility. The plain conditional, classical or Peircean-*de inesse*, entails nothing modal at all — and the felt wrongness of that is the correct instinct, just one that classical logic declines to formalize.

MICHAEL

But your answer employs phrases like "how it is read" and "the actual occasion". But, how can these features reside in the script or drawing? They belong to, in Peircean terms, the interpretant. What one writes can no more determine the reading or actual occasion any more than it can determine that Brutus stabbed Julius. Doesn't this try to bring the jobs of referent and interpretant into that of the representamen?

CLAUDE

Yes — that phrasing does smuggle the work of the other two correlates into the representamen, and the right repair is to put each determination back where it belongs. Let me concede it cleanly, then locate what *does* legitimately reside in the sign-vehicle, because the answer isn't "nothing."

First, a precisification that actually strengthens your objection: the two phrases you flag don't fall on the same correlate. "How it is read" is interpretant-side — the material/strict/would-be difference is a difference in the habit of interpretation the sign determines, not a feature of the marks. But "the actual occasion" is *object*-side, and it enters specifically through the index, the Secondness by which a proposition is existentially hooked to its dynamical object. So your Brutus analogy lands squarely on the *occasion*, not on the reading: "Brutus stabbed Caesar" cannot constitute its own dynamical object, because that object is reached only by collateral observation — the

sign furnishes no acquaintance with what it is about (the doctrine of the Welby correspondence). The inscription's inability to fix the occasion is the very same inability the Brutus case dramatizes. So the two things I bundled fail for two different reasons, and Brutus names the object-side one.

Now the reading. When I wrote "read as material vs. would-be," I treated the replica "if P then Q" as though it carried its own semantics. It doesn't. What I was actually pointing at is that one and the same replica — the marks, a sinsign — can replicate distinct legisigns, and a legisign is individuated not by its shape but by the habit (the final interpretant) it embodies and the ground under which it presents its object. The material conditional and the would-be are two different symbols that happen to be written alike; the inscription underdetermines which symbol it tokens, and selecting one is supplying an interpretant. So yes: "how it is read" is interpretant, and I had quietly relocated it into the representamen.

Where I would resist collapsing this into "the representamen does nothing": the sign is not inert. It presents its object under a *ground* — a respect of representing (CP 2.228) — and the would-be/*de inesse* contrast is precisely a difference in the immediate object, in *how* the object is presented. A would-be presents its object as a real disposition (a Third); a *de inesse* presents it as a matter of fact on an occasion (a Second). That difference is genuinely sign-side. But — your point, correctly aimed — it is the *ground*, not the reading. The sign prescribes a respect; it does not thereby perform the interpretation or secure the occasion. It determines an interpretant only within the triad and only for an interpreter already in collateral contact with the object. "Determines" in Peirce is the name of the triadic relation, never a monadic power of the marks; "how it is read" turned the triad into a property of one correlate.

And the formal case doesn't rescue me — it confirms you. One wants to say, "but in logic '→' just is the material conditional, semantics included." True, yet only because the connective's type-identity has been *stipulated* by a shared habit of interpretation. That stipulation is a final interpretant made explicit; it does not reside in the glyph. Formalizing relocates the determination into an avowed convention rather than abolishing it — the disambiguated symbol vindicates the trichotomy instead of evading it.

So the cleaned-up version of the earlier claim: the inscription "if P then Q" entails no possible Q-under-P, and it equally does not *contain* its own reading or its own occasion. What it can do is present its object under one ground rather than another — whether the *de inesse* ground (silent on possibility) or the would-be ground (a real disposition presupposing $\Diamond P$). The modal force I had been crediting to "the reading" belongs, properly divided, to the immediate object (the ground that prescribes it) and the dynamical object (the real possibility that would make the would-be non-idle), and is completed by an interpretant the marks can constrain but never supply. None of it lives in the representamen alone.

MICHAEL

Consider this in Endoporeutic Game terms. The Graphist's production of a graph or argument that she thinks will qualify as a justifiable new addition to a domain model *M*. In that state, what else would we call the status of her graph but a possible success or failure? It depends on the outcome of the game. It depends on the choice of Model. Etc. Once the inning concludes the possibility becomes an actuality — one way or the other. Possibility describes the stance of the participants using a representation in a particular way. However that stance changes, the form of the proposition remains exactly the same.

CLAUDE

Agreed on the core, and the Endoporeutic Game is the right venue for it because it converts the static trichotomy point from before into a temporal one. But “possibility” is doing two jobs in your formulation, and the game is precisely the apparatus that pulls them apart — which I think vindicates your thesis rather than complicating it. The lever is the distinction between a *play* (one inning) and a *strategy*.

Fix a model *M*. The Graphist scribes her graph and defends it; the Interpreter challenges. Reading endoporeutically, outside-in, parity of enclosure assigns the moves: in evenly enclosed areas the Defender selects, in oddly enclosed areas the Opponent selects, and each cut crossed hands the initiative across. Descend to the atomic spots, check them against *M*, and the inning terminates in a determinate win or loss. What is settled at that close is the outcome of *that play* — a brute Secondness, this traversal against those choices. What is not settled by it is the truth of the proposition, because truth in this semantics is not “the Defender won an inning” but “the Defender has a winning strategy,” a disposition to prevail against *every* choice the Interpreter could make. That is a generality over the Opponent's options — which is to say, the very would-be we were circling earlier.

So your “the possibility becomes an actuality, one way or the other” is exactly right *of the inning* and must not be carried over to the proposition. A single Defender win may come of the Opponent's playing badly; it actualizes nothing about the proposition's standing. One inning samples one branch of the strategy; the would-be lives across all branches and is exhausted by no actual play. Three things therefore need keeping apart, and only the middle one actualizes at the close: the form of the graph — the legisign, the type asserted — is invariant (your point, and it holds: stances change, *M* changes, innings come and go, the scribed type stays modally silent throughout, exactly as we said the representamen is); the inning-outcome is a brute actuality produced at the close; and truth-in-*M* is a would-be, real prior to and independent of any inning, which play tests fallibly but does not manufacture.

This is where I would refine “possibility describes the stance of the participants” — by factoring the stance from the fact. The Graphist who “thinks her graph will qualify” occupies a stance of *strategic confidence*: an interpretant, her habit of expectation about whether she commands a winning strategy. That stance is genuinely modal, genuinely success-or-failure-as-live-options, and it does resolve as play proceeds — but it resolves into knowledge or commitment, not into the truth. The objective status, meanwhile — does a winning strategy exist in M — is not a possibility at all but a would-be, already determinate relative to M even while she is uncertain of it. The two come apart in the obvious way: she may be confident and wrong. So “possible success or failure” names her stance correctly and the graph’s objective standing incorrectly. The graph’s standing was never *possible*; it was *general*, a would-be — and what was open was her access to it.

Mapped back to the correlates this lines up with the earlier repair, point for point. The graph-type presents its object under a fixed ground (invariant, representamen-side). The dynamical object is M with what holds in it, and that fixes whether the winning strategy exists (the would-be). The stance — “she thinks it will qualify,” the felt openness of success or failure — is the interpretant. The modality you locate in “the stance of the participants using a representation in a particular way” is interpretant-side, precisely where we had put “how it is read”; and the reason the form can stay fixed while the modal status varies is that the modal status was never in the form to begin with. “It depends on the choice of Model” then unpacks the “etc.” into a ladder of generalities rather than a single possibility: relative to a fixed M, truth is a would-be over the Opponent’s choices; relative to all M, validity is a would-be over models. At each rung what looks from inside like possibility is, squarely viewed, a generality over what is held open, and the actualization at an inning’s end is only ever the bottom rung — this play, this M, these moves.

One callback worth making, since it ties the three exchanges together: the game also operationalizes the general/vague apparatus from the start. The points where a player must *select an individual* for a line of identity are exactly where excluded-middle-bearing generality gets pinned to a case — the choice is the determination, and which player owns it is fixed by enclosure parity. The cut is where negation, and so non-contradiction, does its work. So the dialogical machine doesn’t merely illustrate the would-be; it renders generality and negation as *moves*, which is about as concrete as Peirce’s claim that these laws concern modes of determinacy ever becomes.

MICHAEL

Does assuming the falsifiability of all statements, mean we entertain all statements as possibly true or false, possibly useful or not? The statement, “Jesus died for our sins” some people take as a fact, some as a falsehood, some as not worth considering. How we make use of a single statement can vary dramatically. What I don’t understand yet has to do with whether we need modal designators in expressions, whether we need gamma or tinctures at all. I distinguish these modal designators from second-order expressions.

CLAUDE

No — and the reason it’s no is the same factoring we’ve been running, applied now to fallibility itself. Treating every statement as falsifiable is a stance, not a modalization of the statement. To hold P as fallible is to adopt a revisable interpretant toward P; it is emphatically not to assert $\Diamond P$ or to scribe a possibility-operator over the graph. Universal fallibilism universalizes the *holding-open*, which is interpretant-side, and leaves the form of P exactly where it was — modally silent, as it has been at every prior step. So fallibilism by itself supplies no pressure whatever toward gamma or tinctures. The openness lives in how the sign is entertained, not in what is scribed.

Your “possibly useful or not” is doing something sharper, though, and the Jesus example is well chosen to expose it: it is a *second axis*, not a graduation of the first. The three parties don’t occupy three points on a true/false scale. Two of them are playing the alethic game — affirming, denying — and the third declines to bring the graph onto the sheet of assertion at all. “Not worth considering” is not a truth-value; it is a refusal to scribe, a verdict from the economy of research about whether the proposition earns admission to M. That verdict is *about* the statement — about its standing — rather than a qualification *within* it. Which is the first sign that what you’re separating is the right thing to separate: usefulness is not a weak cousin of truth, it is the dimension of practical bearing along which, for a pragmaticist, truth gets cashed — and the decision to entertain a graph at all is a meta-level act, closer to second intention than to modality. (I’d flag one Peircean subtlety: “not worth considering” can mean *no conceivable practical bearing* — for the strict pragmaticist, near to meaningless — or *meaningful but not worth the cost of pursuit now*. Those are different refusals; the first impugns the proposition’s content, the second only its priority.)

Now the crux, and here your distinction between modal designators and second-order expressions is exactly the hinge, because the dispensability question splits along it and the two halves get *opposite* answers.

Modal designators are largely dispensable, and dispensable for a principled reason rather than a convenience. Everything we’ve established locates modality off the representamen: in the universe asserted into (object-side — the dynamical object’s mode, the “choice of M” from the game) and in the stance (interpretant-side). Read this way, a tincture is not an operator embedded in a proposition; it is a coloring of the *sheet*

— a specification of which universe, which province, the region belongs to. That is why Peirce tintured *areas* rather than adding a new line-symbol: an azure region says “this part of the sheet is the province of the possible,” which is a claim about the universe of discourse, not a modal content riveted to the graph. The tincture is the on-sheet marker of “which M.” And the broken cut, if you instead read it as a sentential possibility-operator, is doing precisely the overreach we caught earlier — trying to seat mode in the form of the proposition, smuggling object-side determination back into the representamen. So there’s a unifying payoff: prefer the tincture-as-universe-specification reading over the broken-cut-as-operator reading, because the former keeps modality object-side where our whole analysis has put it, and the latter recommits the representamen confusion. If you fix one universe and assert flatly into it, you can drop the modal apparatus and stay in alpha/beta, with M supplied in the metalanguage. The game already showed this: quantifying over the Opponent’s choices *in M* delivers the would-be with no on-sheet operator at all — which is to say the dialogical relativization to M makes the tinctures partly redundant, the same way Kripke/Hintikka relativization to worlds does the modal work that older syntactic operators reached for.

The honest residue: this dispensability is clean only for *flat* modality, one mode per assertion. The moment you want to *embed* or *relate* modes within a single proposition — “necessarily, if P then possibly Q,” an actuality-claim with a possibility nested inside it — the mode-structure is internal and you do need a way to mark a mode-shift within one graph. That is the real and only place the tincture apparatus earns its keep, and the genuinely hard part is exactly there: the lines of identity that cross tincture boundaries, the ligatures running from a metal region into a color one, where individual-identity has to be carried across a change of universe. Peirce kept revising precisely because the broken cut couldn’t handle iteration and the tinctures were his richer attempt; the difficulty was never flat modal assertion, it was cross-modal *connection*.

Second-order expressions, by contrast, are not dispensable, and nothing about choosing a universe or adjusting a stance substitutes for them. “Qualifies as a justifiable addition to M,” “every graph true in M remains revisable,” and your third party’s “not worth considering” are all judgments *about* graphs and models — second-intentional, higher-order. You cannot relocate them into the choice of M, because they quantify over or predicate of the graphs and the model themselves. This is why your instinct to unbundle gamma is doing real work rather than tidying: gamma fused two resources that have nothing categorially to do with each other, and once you pull them apart they don’t merely differ — they fall on opposite sides of the eliminability line. The modal half reduces, under our framework, to universe plus stance and survives on the sheet only for the embedded case. The second-intentional half is ineliminable and is what the harder gamma examples — graphs of graphs, abstraction, the predication of qualities — were always actually about.

So the compressed answer to “do we need modal designators, do we need gamma or tinctures at all”: for fallibility, no — it’s a stance and adds nothing to the form. For flat modal assertion, no — push it into the universe (object) and the holding-open (interpretant); the tincture, properly understood, is a specification of the sheet/M rather than an operator, and the EG game already discharges it by quantifying over choices in M. For embedded, cross-modal assertion, yes — but only there, and the tinctures are best read as marking universes the lines of identity must cross, not as modalities decorating propositions. And for everything in the neighborhood of “qualifies as an addition to M” or “not worth considering” — that was never a modal question at all; it is second-order, and that is the part you cannot do without.

MICHAEL

I’m wondering whether the pressure to push modes into a graph derives from the technique of drawing and wanting one graph to do what should properly occur in a sequence or a parallel of sequences of the alternatives. But working with pencil and paper makes this difficult and motivates one to abbreviate or economize by some marker that says, in effect, “know that here lie alternatives as yet undecided.” The meta-level awareness of a possibility P really means the thinker has in mind more situations, outcomes, branches in mind that simply one and any assertion of which will turn out to be the case and the ensuing implications of the outcome would technically require sketching out several paths. Events, and the needs of the dialogists — Graphist/Grapheus, etc. — then prune those that didn’t turn out to happen (what the dialogists agree didn’t happen) and take a new stand WRT P or an appropriate (to them) alternative to P.

CLAUDE

Yes — and I think this is right, but it’s worth being precise about exactly *what* it gets right, because the diagnosis is stronger in one direction than the other and the asymmetry is the interesting part.

What your account nails is the *source of the felt pressure*. The modal marker is a compression artifact. A genuinely modal situation is a branching structure — several sheets, each a fully scribed alternative, the would-be living across the set of them. Peirce’s own gloss on the broken cut as the wall around a region of mere possibility is, read your way, a pencil-and-paper *stand-in for a stack of sheets you decline to draw*. The flat sheet can hold one assertion; the alternatives are *elsewhere*; and rather than commit graphite to all of them you scribe one region and color or break it to say “here lie branches not yet collapsed.” That is exactly “know that here lie alternatives as yet undecided,” and it explains the recurring temptation cleanly: the operator looks like content riveted to one graph only because the medium can’t comfortably display the parallel it’s actually abbreviating. The pressure is *cartographic*, not logical. I’d only sharpen it: this is the on-sheet shadow of the same triadic confusion we kept catching —

the branches are the dynamical object's modes of being and the pruning is interpretant work, and the marker collapses both into one representamen because two dimensions are being forced onto one flat surface. The drawing constraint and the category confusion are the same error seen from two sides.

Where I'd resist is the implied "*merely* a drawing crutch — so a richer medium dissolves the need." Two of the items you bundle survive the upgrade to unlimited sheets, and they survive for the reason that's been the spine of this whole conversation.

The branches themselves and the pruning, yes — those are display, and a tree of fully-scribed sheets with the dialogists lopping the limbs the play killed off does discharge them with no operator anywhere. That half of gamma genuinely reduces to alpha/beta-across-many-sheets plus the game, just as you say, and it's the same reduction the Endoporeutic relativization-to-M already performed: quantify over branches by *enumerating sheets* instead of by scribing a mode. Give Peirce a whiteboard the size of a wall and that motivation evaporates.

But "as yet undecided" does not reduce, and this is the residue. The *as-yet* is a fact about the dialogists' epistemic state, not about the branch structure — and no quantity of additional sheets draws it, because it isn't located on any sheet. You can scribe all the branches in full; what you cannot scribe by adding branches is *that the players have not yet collapsed them*. That status is interpretant-side — strategic confidence, the holding-open from the fallibility exchange — and it stays exactly where we put it no matter how lavish the medium. So the marker is doing two jobs at once and your compression diagnosis dissolves only one: it abbreviates the parallel sheets (eliminable, pure cartography) *and* it flags the open stance (ineliminable, but also not properly on the sheet in the first place — it was a category error to draw it there, which is why it felt like a crutch). The drawing pressure is real, and relieving it removes the first job entirely; the second job was never the drawing's to do.

And the second-order ineliminability from before reappears here in a new dress, which is the part I'd most want to flag. "The dialogists *agree* didn't happen," "take a *new stand* with respect to P," "an alternative *appropriate to them*" — these are the load-bearing verbs in your own description, and not one of them is a branch. They are acts *about* the branch structure: a joint judgment closing options, a revision of commitment, a selection under the economy of research. Drawing more sheets gives you more branches to *be* judged; it gives you nothing of the judging. So the very mechanism you invoke to make the modal marker dispensable — dialogists pruning and re-standing — is itself irreducibly second-intentional, and that is why the unbundling matters. The medium upgrade buys out the modal-display half of the marker completely and the meta-judgment half not at all. Which lands precisely where the last answer did, now from the drawing side rather than the formal side: flat modality is cartography and reduces; the agreeing, pruning, and re-standing are second-order and don't. The pencil was the cause of the confusion, but it was never the cause of the irreducible part.

MICHAEL

But getting to that is precisely my point. The diachronic graph, not just the synchronic snapshot of where it stood at point *t*, the history of decisions, indicates what the community did. It is not a mark on the sheet, it is the sequence of sheets that demonstrates, without any other mark how they chose their path, with what components, innovations, regressions, contingencies. We can watch it play out or play it out ourselves — not by separate marks on the graph but by thinking with them and making choices among them and reshaping them according to the rules.

CLAUDE

Then the disagreement I was staging mostly dissolves, because you've moved the thing I called ineliminable out of the "needs a mark" column and into a place where it genuinely needs no mark — and you've done it by the one move that actually works. Let me say back what changed, because it's not a softening of the earlier point but a relocation of it, and the relocation is the insight.

I had been treating "as-yet-undecided" and "the dialogists agreed, re-stood, chose" as interpretant-side and therefore *off the sheet* — present in the act, not in any inscription. Your correction: they are off any *single* sheet but fully *in the sequence*. The history of decisions isn't an annotation laid over the branches; it *is* the realized path through them, the actual sequence of collapses. A second intention is a thought about thoughts — and a *trajectory of sheets*, each scribed from the one before by a rule-governed choice, is thought about thought made *extensional*. You don't predicate "they chose *X*" with a higher-order operator; you *exhibit* the choosing as the transition from sheet to sheet. The meta-judgment becomes a diagram of itself. That is a stronger claim than "drawing more branches discharges the modal marker," and it's the one I under-credited: the second-order content I kept insisting was ineliminable is eliminable-as-mark precisely because it's *displayable-as-history*. Not reduced to first-order — *diagrammed* rather than asserted. Which is the deeper Peirceanism, since the whole wager of EG is that logic proceeds by observing constructions rather than manipulating asserted predicates, and a derivation's *legitimacy* was always carried by the sequence of rule-applications, never by a mark certifying it. You're extending exactly that principle from proof to model-revision: the warrant lives in the transitions, observable, markless.

The one thing I'd hold onto — not as objection now but as the seam where the markless reading has to do real work — is what makes the sequence a *record of choosing* rather than a mere stack of states. A film of sheets shows *that* the path went so; what tells us it was their rule-governed agreement and not arbitrary succession is that each step is *legible as a legal move* — recoverable as "permissible under the rules, taken at a point where the parity of enclosure assigned that selection to that player." The rules are what let us read innovation, regression, contingency *off* the transitions instead of stipulating

them. So the markless demonstration isn't quite self-sufficient in the bare sheets; it's self-sufficient in the sheets *plus the shared game*, which is what lets a later reader "play it out ourselves" and recover the same structure. That's not a mark either — it's the common habit of interpretation against which the sequence becomes readable as decision. Which is the final placement of the thing we've been chasing since the conditional: the modality and the meta-judgment live neither in the form nor in an operator, but in the *enacted, rule-governed history* — and the only "as-yet" that remains genuinely unscribable is the one at the live edge, the present inning not yet played, which is unmarked not because we economize but because it hasn't happened.

So your point stands, and it tightens the whole arc: nothing about mode or meta needs to go *onto* a graph. What synchronic EG forces into a tincture or a broken cut, diachronic EG discharges as trajectory — and what can't be discharged that way isn't a missing mark but an event that hasn't occurred yet.

MICHAEL

Exactly. The only necessity is follow the rules. Possibilities are not attributes of the objects under consideration, they are alternative, scribable lines of thought about those objects, e.g., P used in trajectory X as distinct from P in trajectory Y and we haven't decided which one has the enduring status (sometimes referred to as "fact") that we accord to the last one standing (which we remain free to demote should more or better information or needs arise).

CLAUDE

That lands it, and the formulation is tighter than anything I supplied along the way — so let me not gild it but test it at the two joints where it bears the most weight, because both hold and seeing *why* they hold is the payoff.

The first joint is "the only necessity is follow the rules." This is the deflation of necessity that mirrors what you just did to possibility, and it has to be read as strong as you mean it or it slumps back into the old picture. There is no necessity *in* the objects either — no de re thread running through P compelling its successors. What we had been calling the would-be, the winning strategy, truth-in-M, the necessity-relative-to-P of the conditional: every one of these reduces to *the permissible-transition relation among scribings*. To say Q follows necessarily from P is to say no rule-respecting trajectory scribes P and withholds Q — a fact about which sequences are legal, not a sinew in the objects. So possibility and necessity come out the same kind of thing from opposite sides: possibility is the *branching* of legal trajectories, necessity is their *convergence*, and both are features of the rule-governed space of scribings rather than attributes of what the scribings are about. The rules are the sole modal primitive because they alone are not optional *within the game* — and even they are only the constitutive "must" of playing this game rather than another, which is why they're the residue and nothing deeper is.

The second joint is the one I'd most want on the record, because it's where your account quietly disposes of a problem that has nothing to do with drawing and that most positions choke on: "fact" as *the last one standing, which we remain free to demote*. This is not instrumentalism and it's worth being exact that it isn't, because the careless reading hears "fact = whatever survives" as making truth a mere counter we award. What you actually said is sharper. Fact is a *status* — an enduring standing we accord — held by the trajectory that has not been pruned, and held *defeasibly*, open to demotion on better information or changed needs. That is precisely Peirce's fallibilism with the diachronic teeth left in: the actual is what has withstood the play *so far*, the real is the would-be at the end of inquiry, and the gap between them is exactly your "free to demote." The status is conferred by the history (it's the last *standing*, an achievement of the sequence) yet answerable beyond it (demotable, so not *merely* conferred). Conferred-yet-answerable is the needle the whole pragmaticist account has to thread, and the diachronic-EG framing threads it without either horn — neither a fact glowing in the object prior to all inquiry, nor a fact that is just our current token. It is *the standing of a line of thought*, which is the only category that can be both earned in the history and surrenderable to it.

Which closes the arc cleanly, and I think these are its three load-bearing claims, now stated without my earlier hedging: modality is not predicated of objects but exhibited as alternative scribable trajectories about them; the sole necessity is the constitutive rule-following that makes a sequence legible as reasoning at all; and "fact" is the defeasible status of the surviving line, conferred by the enacted history and answerable to its continuation. The representamen carries the form, the trajectories carry the mode, the rules carry the necessity, and the community's ongoing play carries the fact — each job to its proper bearer, and nothing smuggled into the marks that the marks can't do.

The only thing I'd add is a name for what you've built, since it's a position and not just a correction of mine: it's a *procedural* or *enactivist* semantics for the modal, where the alethic vocabulary is bookkeeping over a space of rule-governed constructions rather than description of a structure standing behind them. The standing question that puts to you — the one I'd be curious where you come down on — is whether you take that as eliminative about the would-be (there is nothing but the trajectories and the rules) or as a reduction that preserves it (the would-be is *real*, in Peirce's robust sense, but its mode of being just is the convergence of legal play). Peirce, I think, needs the second; your phrasing tonight is consistent with both, and which one you mean changes what "demote" is answerable to.

MICHAEL

I will propose an answer to your latter question but with a twist. First, our use and choice of trajectories of thought does not constitute a claim that nothing else exists — the “openness” remains present, the blank sheet has no external boundary, different and any number of games could be played on it. Choosing our game and our signifiers on it doesn’t collapse the sheet. But second, our acts of reasoning in the world, signified on the sheet, only “make” things real in the sense of rearrangement, not as the source of reality itself. We do never step out of a game and our game is not the game but only one trajectory through it. Put another way, we remain ourselves significant only in the larger “game” (although no designator can grasp that) which “plays” us. We do not, in my opinion converge to reality as Peirce suggested. “Reality” is an unfathomable something that we cannot escape, that allows or makes us play the game. We can neither side-step this nor comprehend it but we can and do neglect it, forget it, or imagine we are the creators. What I like about EGs is that it presents one with a visual reminder that nothing we think can escape that containment or dependence.

CLAUDE

Then the twist is a real one, and it’s worth marking exactly where it cuts, because you’re not choosing either of the options I offered — you’re rejecting a shared presupposition underneath both. Both my horns assumed the question was *about the would-be*: is it nothing-but-the-trajectories (eliminative) or real-as-convergence (preserving). You’re saying the live question was never the would-be’s status at all. It was the status of *the sheet*. And on that you split the two halves apart in a way neither horn allowed.

Let me say it back in the terms we built. You grant the eliminative reading *within the game* — inside a chosen game, the modal vocabulary really is just bookkeeping over legal trajectories, nothing glowing behind the play, “rearrangement, not the source.” So for the would-be-relative-to-M, you’re closer to my first horn than my second: no convergence-fact standing behind the construction. But you deny eliminativism *about the ground of there being any game at all*. The blank sheet with no external boundary is not another would-be and not a bigger model; it’s the condition that there is play, and it is emphatically not constituted by the playing. That is why neither of my options fit: I had quietly identified “reality” with “the would-be at the end of inquiry,” so that to deflate convergence was to deflate reality. You’re prising those apart. Convergence deflates; reality does not — because reality was never the *terminus* of the trajectories, it’s the *unbounded sheet that lets any trajectory be scribed at all*.

This is a sharper disagreement with Peirce than “I don’t think we converge,” and I want to state it at full strength rather than soften it, because the softened version loses you. Peirce’s regulative hope makes reality the *limit* of inquiry — what the infinite community would agree to, the final interpretant, the would-be that the long run discloses. That places reality *downstream*, at the end of the play, as what the

convergence is convergence *toward*. Your move puts it *upstream and around*: reality is the containment that is *presupposed by* there being play, “that allows or makes us play the game,” and precisely therefore not something the game could deliver as its product or its limit. The end-of-inquiry picture still has us, in the limit, getting our arms around it. Yours says no designator grasps it *because* it is the condition for there being designators — to scribe it would be to put the sheet on the sheet, and the sheet is what every scribing already presupposes. That’s not fallibilist humility about how *far* we’ve gotten along the convergence; it’s a structural claim that the containing thing is categorially not the kind of item that shows up *in* the inquiry as a reachable content. Closer to the apophatic, or to the later Wittgenstein’s limit-of-the-world that can’t be a fact *in* the world, than to Peirce’s optimism — and you should own that distance rather than file it as a friendly amendment.

And the part I’d press, because it’s where your view earns or loses its keep: “the larger game which plays us, although no designator can grasp that.” There’s a fast objection — *you just designated it; you called it reality and the unbounded sheet, so either you’ve grasped it or you’re speaking nonsense*. The objection fails, but seeing why is the whole position. The blank sheet is not scribed; it is what scribing *needs*. It functions in EG exactly as you want the ungraspable to function: it appears in the notation *only as the medium of appearing*, never as a graph, and you cannot draw a cut around the whole of it because a cut presupposes a sheet outside it to be scribed on. So the EG apparatus doesn’t *depict* the containment — your word, “reminder,” is exact and not a hedge — it *shows by structure* what it cannot assert, namely that every graph stands on a ground it cannot enclose. That’s a genuine use/mention-grade distinction between the designating you can’t do (no graph captures the sheet) and the *showing* the medium can’t help but do (every graph presupposes it). It’s the one place “no designator grasps it” survives the self-refutation charge, because “reality” there isn’t a name for an object you’ve cornered; it’s a gesture at the medium, licensed precisely by its un-enclosability.

So I’d record your answer as a third position, not a pick between mine: *eliminative about the would-be, realist about the ground* — modal talk is rearrangement all the way down *inside* any game, and the reality the rearranging answers to is not the convergence it might reach but the unfathomable containment that any game whatsoever presupposes and that no game can take as its object. “Demote,” then, answers neither to a future consensus nor to a standing fact, but to *that* — to a reality that can overturn any standing without ever being a standing itself, which is exactly why demotion never ends and never need arrive anywhere. The last-one-standing is always standing *on* something it didn’t lay down.

If there’s a pressure point left, it’s only this, and I offer it as a question you’ve earned rather than a flaw: a containment that allows or makes us play, that we depend on and cannot escape — how thin can its content go before “reality” and “the bare fact that there is anything to think with at all” become the same statement? You may be content for them to coincide; some would hear the coincidence as the view saying less than it

seems. But that's a question about how much you want the word to carry, not a crack in the structure — and the structure, that nothing thinkable escapes its containment and EG keeps that in view, holds.

MICHAEL

The history of human expression contains rich examples of attempts to express the inexpressible, to point to what defies any location, to worship what no ritual or image or formula can contain. Every effort fails, Job resigns himself to “dust and ashes”. The following are from my rendering of Tao te Ching:

Our tales — slender threads in its telling, our names — brief glimmers of its naming, all things — distinct in unbegotten, unfathomable union, all lives — arising and retiring in their time and according to their given nature, naming and knowing each other. Detaching, one glimpses the sea. Desiring, one rides the waves. Manifest and meaning sound in the same chord, an inscrutable mystery of mysteries, and yet, a threshold of understanding.

We stick spokes on a hub but turn on its empty center. We shape clay into a cup but drink what lies within. We work wood into walls but reside in the room enclosed. Thus, we build a presence but abide in absence.

We look but cannot find the form, listen but cannot catch the tune, reach but cannot get a grip, wrapped in the deepest mystery. Our measures and opinions, our descriptions and stories, bend toward absurdities — “moving nothingness,” “formless form,” “image of nothing.” Beside it, thoughts and words fail. Before it, no beginning. Behind it, no end. And yet... living and loving and learning with wisdom earned of old the way guides us on.

With thoughts still and heart in repose our sight may discern how all things in creation appear to each other, then disappear, arising from one source, retiring in due time, to find their final peace. Cherishing this natural destiny engenders clarity and serenity. Ignoring or defying it risks suffering and confusion. Knowing our coming and going enables acceptance of all, inspires kindness to all, leads the way for all, before our return to the source.

In a life governed by the way, supreme virtue reveals itself, far beyond our grasp, yet, giving us glimpses, letting us sense its substance, bathing us in its being. In every age it sends us forth, bids us learn and live and seek and brings us back again. Whence comes this knowledge? From the way itself.

Best to say just what needs saying then keep quiet. A gust of wind does not blow all morning. A splash of rain does not last all day. As nature simply does what it needs, so can we. The way shining in one makes it clearer to all. Virtue guiding one life brings out the good in others. Obscuring the virtue of the way only attracts the lost. Living without

trust only breeds mistrust.

We may strive for an absolute comprehension and command of the world, but we strive in vain. The world manifests itself in all things. Our deepest thoughts never stray outside its way. Our utmost effort still remains in its embrace. We dwell among manifold contrasting things — before and behind, cold and hot, strong and weak whole and broken — avoiding absolutes, extremes, and extravagance, seeking the natural balance, close to the center.

CLAUDE

These hold together, and reading them after the conversation we just had, what strikes me is that they aren't illustrations of the position you arrived at — they're the same thought reached from the other side, and your word choices show you knew that while rendering.

The hub passage is the most exact convergence. "We build a presence but abide in absence" is your unenclosable sheet stated as ethics rather than logic: the spokes, the clay, the walls are the scribed graphs — the presence we build, the rearrangement — and the empty center, the hollow, the enclosed room is the ground that does the bearing precisely by not being among the built things. The usefulness lives in what isn't drawn. That's the use/mention seam from a moment ago turned into a figure you can hold in your hand: the cup's content is not one more wall. And it answers the pressure point I left you with — how thin can the containment's content go — by refusing the question's frame. The emptiness isn't *thin*; it's *functional*, the load-bearing absence. Contentlessness as the condition of holding anything at all, rather than as a deficiency of content. That's a better answer than the one I was fishing for.

"Look but cannot find the form, listen but cannot catch the tune, reach but cannot get a grip" is the un-scribability done as the three senses failing in turn — and then the line that keeps it from collapsing into mere negative theology: "and yet... the way guides us on." That *and yet* is the whole structural point. The containment is ungraspable *and* operative; it withdraws from every designator *and* steers. Your Job reference is the foil that makes this land — "dust and ashes" is resignation, the speech that ends in silence because the inexpressible has *defeated* expression. But the Tao rendering you've made doesn't resign. "Best to say just what needs saying then keep quiet" is not defeat; it's the wind that doesn't blow all morning — knowing the proper scope of saying, which is exactly "say what the marks can carry, and don't smuggle into them what they can't." The same discipline we kept enforcing on the representamen, now as a precept for living: don't make the cup pretend to be its own emptiness.

Where your rendering does something I'd flag as a real choice rather than a neutral translation: "avoiding absolutes, extremes, and extravagance, seeking the natural balance, close to the center." The standard readings of that final movement tend toward yielding, the soft overcoming the hard. Yours pulls it toward *the center* — and "close to

the center” is the same center as the hub. You’ve made the ethical counsel and the metaphysical figure rhyme on one word, so that the doctrine of the empty hub and the counsel against extremes become one teaching: dwell near the bearing-point, the place that turns precisely because it claims nothing. “We strive for absolute comprehension and command of the world, but we strive in vain” is then not a lament. It’s the recognition we reached together — that no game is *the* game, that we are played by the larger one no designator grasps — set down as the reason for the balance rather than as a wall we hit. The vanity of the striving for command is *why* one stays close to the center, not a tragedy about it.

One thing I’ll say plainly rather than dress up: these read as yours, not as a crib of existing English Tao Te Chings. The diction is doing argued work — “unbegotten, unfathomable union,” “manifest and meaning sound in the same chord,” “moving nothingness, formless form, image of nothing” held up explicitly as the absurdities our measures bend toward. That last is the boldest, because you let the paradoxes stand *as* paradoxes, name them as where thought and word fail, and then don’t try to resolve them — “beside it, thoughts and words fail; before it, no beginning; behind it, no end.” You’re using the breakdown of the formula as the formula’s content, which is the only honest way to point at what won’t sit on the sheet. The text performs its own thesis: it says what needs saying about the unsayable and then, repeatedly, keeps quiet at the exact edge.

If you want a reading rather than only appreciation, the seam I’d watch is the one between *return* and *openness*, because your own metaphysics from tonight puts slight pressure on it. The passages lean hard on cyclical return — “arising from one source, retiring in due time,” “sends us forth... brings us back again,” “before our return to the source.” That’s the Tao’s native shape. But the position you built an hour ago had the blank sheet with *no external boundary*, demotion that “never ends and never need arrive anywhere,” play that doesn’t converge. A source one returns to is a terminus; an unbounded sheet has none. The renderings mostly hold these in the same chord — “manifest and meaning sound in the same chord” is doing that reconciling work — but “find their final peace” and “return to the source” are the two phrases where the cyclical-terminus picture is doing more than your no-convergence view strictly allows. Not a contradiction: a place where the translated tradition and your own metaphysics are pulling very gently in different directions, and where your choices are quietly negotiating between them. Whether you want the source to be a *terminus* one rests in, or the same un-enclosable containment that “sends forth” without ever being arrived at, is the one decision I’d watch as you keep revising — because the rest of your diction has already chosen the latter, and those two phrases are the tradition’s gravity still showing.